Polar Form, De Moivre's Theorem & Roots of Unity

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Polar and Euler Forms**

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

1. Expand $(\cos\theta + i\sin\theta)^3$ via Binomial Theorem: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. 2. By De Moivre's Theorem, this equals $\cos 3\theta + i\sin 3\theta$. Equate real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$. Substitute $\sin^2\theta = 1 - \cos^2\theta$.

2 **[Foundational]** • **Target: De Moivre's Theorem for Powers**

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

1. Step formula: $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h(x_n + y_n)$ with $h = 0.1$, $x_0 = 0$, $y_0 = 1$. Step 1 (to $x_1 = 0.1$): $y_1 = y_0 + 0.1(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.1$. Step 2 (to $x_2 = 0.2$): $y_2 = y_1 + 0.1(x_1 + y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22$. Euler's approximation is $y(0.2) \approx 1.22$. 2. Exact analytical solution: Rewrite as $\frac{dy}{dx} - y = x$. Integrating factor $\mu = e^{-x}$.

3 **[Foundational]** • **Target: Roots of Complex Numbers**

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.

To find the x-coordinates where the graph intersects the x-axis, we need to find the roots of the function, i.e., where $f(x) = 0$. We set the function equal to zero: $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 6 and add to -5. These numbers are -2 and -3. So, the equation can be factored as: $(x - 2)(x - 3) = 0$. For the product of two factors to be zero, at least one of the factors must be zero. Therefore, $x - 2 = 0$ or $x - 3 = 0$. Solvi.

4 **[Foundational]** • **Target: Roots of Unity & Factorization**

Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

Let the function be $f(x) = ax^2 + bx + c$. We are given three points that lie on the graph of $f(x)$: $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Substituting these points into the function, we get a system of three linear equations with three unknowns a, b, and c: 1. For $(-1, 12)$: $a(-1)^2 + b(-1) + c = 12 \Rightarrow a - b + c = 12$ 2. For $(1, 2)$: $a(1)^2 + b(1) + c = 2 \Rightarrow a + b + c = 2$ 3. For $(2, 9)$: $a(2)^2 + b(2) + c = 9 \Rightarrow 4a + 2b + c = 9$ We can solve this system of equations. Subtract.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Polar and Euler Forms

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

6 [Foundational] • Target: De Moivre's Theorem for Powers

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2\rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1\rangle$ are not scalar multiples ($1/2eq - 1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1-2) - \mathbf{j}(1-4) + \mathbf{k}(1-(-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1).$ Magnitude.

7 [Foundational] • Target: Roots of Complex Numbers

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

The quadratic equation $ax^2 + bx + c = 0$ has exactly one real root (i.e., the parabola $y = ax^2 + bx + c$ intersects the x-axis at exactly one point) if and only if its discriminant, $\Delta = b^2 - 4ac$, is equal to zero. In the given equation, $x^2 - kx + 8 = 0$, we have $a = 1$, $b = -k$, and $c = 8$. Therefore, the discriminant is $\Delta = (-k)^2 - 4(1)(8) = k^2 - 32$. For the parabola to intersect the x-axis at exactly one point, we must have $\Delta = 0$. So, $k^2 - 32 = 0$. This gives $k^2 = 32$, which m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 **[Intermediate]** • **Target: Roots of Unity & Factorization**

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n \to \infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x \, dx = [e^x]_0^1 = e^1 - e^0 = e - 1.$ (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1} \, dx$. Let $dv = e^x \, dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1} e^x \, dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}.$ ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e - 1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e - 2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

9 **[Intermediate]** • **Target: Polar and Euler Forms**

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

(a) State distribution after 2 weeks is $\mathbf{p}_2 = T^2\mathbf{p}_0$. With $\mathbf{p}_0 = (1; 0)$: $T^2 = (0.7, 0.2; 0.3, 0.8)(0.7, 0.2; 0.3, 0.8) = (0.49 + 0.06, 0.14 + 0.16; 0.21 + 0.24, 0.06 + 0.64) = (0.55, 0.30; 0.45, 0.70).$ Thus $\mathbf{p}_2 = (0.55; 0.45)$, so the probability of being with Brand B after 2 weeks is 0.45 (45%). (b) Steady-state vector $\mathbf{s} = (s_1; s_2)$ satisfies $(T - I)\mathbf{s} = \mathbf{0}$.

10 **[Intermediate]** • **Target: De Moivre's Theorem for Powers**

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

Base Step: For $n = 1$: $5^1 - 1 = 4 = 4(1).$ Since 4 is divisible by 4, the statement $P(1)$ is true. Inductive Hypothesis: Assume $P(k)$ is true for some positive integer $k \geq 1$, meaning: $5^k - 1 = 4m$ for some integer $m \in \mathbb{Z} \Rightarrow 5^k = 4m + 1.$ Inductive Step: Consider $n = k + 1$: $5^{k+1} - 1 = 5 \cdot 5^k - 1.$ Substitute the inductive hypothesis $5^k = 4m + 1$: $5(4m + 1) - 1 = 20m + 5 - 1 = 20m + 4 = 4(5m + 1).$ Since $m \in \mathbb{Z}$, $5m + 1$ is an integer, so.

11 **[Intermediate]** • **Target: Roots of Complex Numbers**

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2 - 3x + 4}{x - 1}$.

To find the equation of the oblique asymptote, divide the numerator by the denominator using polynomial long division or synthetic division: $2x^2 - 3x + 4, = 2x(x - 1) - x + 4;, = 2x(x - 1) - 1(x - 1) + 3;, = (2x - 1)(x - 1) + 3$ Dividing through by $x - 1$ yields: $f(x) = 2x - 1 + \frac{3}{x - 1}$ As $x \to \pm\infty$, the remainder term $\frac{3}{x - 1} \to 0$. Therefore, the linear quotient gives the oblique asymptote: $y = 2x - 1$ - $y = 2x - 3$ arises from mistakenly dividing each of the f.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Roots of Unity & Factorization

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

1. Substitute $x = 2\tan\theta \Rightarrow dx = 2\sec^2\theta\,d\theta$. Then $4 + x^2 = 4 + 4\tan^2\theta = 4\sec^2\theta$, so: $(4+x^2)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$. 2. Substitute into the integral: $\int \frac{2\sec^2\theta}{8\sec^3\theta}\,d\theta = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$. 3. Express $\sin\theta$ in terms of x: From $\tan\theta = \frac{x}{2}$, construct a right triangle with opposite side x, adjacen.

13 [Intermediate] • Target: Polar and Euler Forms

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

1. Sort all edges in non-decreasing order of weight: BC (1), AC (2), DE (2), EF (3), AB (4), BD (5), DF (6), CD (8), CE (10). 2. Apply Kruskal's algorithm (selecting $V - 1 = 6 - 1 = 5$ edges without cycles): - Add BC (weight 1) [Tree components: {B, C}] - Add AC (weight 2) [Merges A: {A, B, C}] - Add DE (weight 2) [Component: {D, E}] - Add EF (weight 3) [Merges F: {D, E, F}] - Consider AB (weight 4): A and B are already connected in {A, B, C} (would create cycle AB-BC-CA), so SKIP AB. - Add BD (weight 5): Connects.

14 [Intermediate] • Target: De Moivre's Theorem for Powers

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

First, we simplify the function $f(x)$. Using the double angle identity for sine, $\sin(2x) = 2\sin(x)\cos(x)$. Therefore, $f(x) = \frac{2\sin(x)\cos(x)}{\cos(x)}$. For $\cos(x) eq 0$, we can cancel the $\cos(x)$ terms, so $f(x) = 2\sin(x)$. Now we need to evaluate the definite integral: $\int_0^{\pi/2} 2\sin(x)dx$. The antiderivative of $2\sin(x)$ is $-2\cos(x)$. Evaluating the definite integral using the Fundamental Theorem of Calculus: $[-2\cos(x)]_0^{\pi/2} = -2\cos(\pi/2) - (-2\cos(0))$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.03 Markov Chains (AI HL)

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

1. Compute the cross product via determinant: $\mathbf{a} \times \mathbf{b} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2, -1, 3; 1, 4, -2| = \mathbf{i}[(-1)(-2) - 3(4)] - \mathbf{j}[2(-2) - 3(1)] + \mathbf{k}[2(4) - (-1)(1)]$ $= \mathbf{i}(2 - 12) - \mathbf{j}(-4 - 3) + \mathbf{k}(8 + 1) = -10\mathbf{i} + 7\mathbf{j} + 9\mathbf{k} = (-10; 7; 9)$. 2. The area of the triangle is half the magnitude of the cross product.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 1.01 Proof by Mathematical Induction

Using the binomial series for $(1-x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x)=\arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1+u)^p = 1 + pu + \frac{p(p-1)}{2!}u^2 + \dots$. Here $u=-x^2$ and $p=-1/2$: $(1-x^2)^{-1/2} = 1 + (-\frac{1}{2})(-x^2) + \frac{(-1/2)(-3/2)}{2}(-x^2)^2 + O(x^6) = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + O(x^6)$. 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, integrate term-by-term: $\arcsin x = \int_0^x \left(1 + \frac{1}{2}t^2 + \frac{3}{8}t^4 + \dots\right)dt = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \dots$ 3. The binomial series converge.

17 [Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A survey was conducted on the number of hours students spent studying per week. The results are summarized in the histogram below. The total number of students surveyed is 120. Let x represent the number of students who studied for more than 3 hours but at most 4 hours. Find the value of x.

The histogram shows the frequency distribution of study hours. The total number of students surveyed is 120. The number of students in each interval is represented by the height of the bars multiplied by the width of the interval. Interval (0, 1]: Frequency = 15 Interval (1, 2]: Frequency = 25 Interval (2, 3]: Frequency = 30 Interval (3, 4]: Frequency = x Interval (4, 5]: Frequency = 20 The sum of the frequencies must equal the total number of students: $15 + 25 + 30 + x + 20 = 120\ 90 + x = 120\ x = 120 - 90$.

18 [Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 1.04 Partial Fractions

Decompose the rational expression $\frac{5x^2-3x+2}{(x-1)(x^2+1)}$ into partial fractions.

1. Set up the partial fraction template with irreducible quadratic: $\frac{5x^2-3x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$. 2. Multiply through by $(x-1)(x^2+1)$: $5x^2-3x+2 = A(x^2+1) + (Bx+C)(x-1)$. 3. Substitute $x=1$ (cover-up): $5(1)^2 - 3(1) + 2 = A(1+1) + 0 \Rightarrow 4 = 2A \Rightarrow A = 2$. 4. Compare coefficients: Coefficient of x^2: $A + B = 5 \Rightarrow 2 + B = 5 \Rightarrow B = 3$. Constant term: $A - C = 2 \Rightarrow 2 - C = 2 \Rightarrow C = 0$. D.

19 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

The derivative of the function $f(x) = 3x^2 - 5x + 2$ is given by $f'(x)$. Find the value of $f'(2)$.

To find the derivative of the function $f(x) = 3x^2 - 5x + 2$, we use the power rule for differentiation, which states that $\frac{d}{dx}(ax^n) = nax^{n-1}$. Applying this rule to each term: 1. The derivative of $3x^2$ is $2 \cdot 3x^{2-1} = 6x$. 2. The derivative of $-5x$ is $1 \cdot (-5)x^{1-1} = -5x^0 = -5$. 3. The derivative of a constant, $+2$, is 0. So, the derivative of $f(x)$ is $f'(x) = 6x - 5$. Now, we need to find the value of $f'(2)$. We substitute $x = 2$ into the expression for $f'(x)$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$ and the variance is given by $Var(X) = np(1 - p)$. We are given that $n = 10$ and $E(X) = 4$. Using the formula for the mean, we have $10p = 4$. Solving for p, we get $p = \frac{4}{10} = 0.4$. Now we can find the variance using the formula $Var(X) = np(1 - p)$. Substituting the values of n and p, we get $Var(X) = 10 \times 0.4 \times (1 - 0.4)$. $Var(X) = 10 \times 0.4 \times 0.6$. $Var(X) = 4 \times 0.6$. $Var(X) = 2.4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Polar and Euler Forms

Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$ ✓

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$

$r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$. In Quadrant III ($x < 0, y < 0$), $\theta = -\pi + \arctan(\sqrt{3}) = -\pi + \frac{\pi}{3} = -\frac{2\pi}{3}$. Thus $z = 4e^{-i2\pi/3}$.

22 [Foundational] • Target: Polar and Euler Forms

Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$ ✓

b. $3\sqrt{2} - 3\sqrt{2}i$

c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

$x = 6\cos(3\pi/4) = 6(-\sqrt{2}/2) = -3\sqrt{2}$. $y = 6\sin(3\pi/4) = 6(\sqrt{2}/2) = 3\sqrt{2}$.

23 [Foundational] • Target: Polar and Euler Forms

If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.

a. $\frac{3}{2}e^{i\pi/6}$ ✓

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

$\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1 - \theta_2)} = \frac{3}{2}e^{i(\pi/3 - \pi/6)} = \frac{3}{2}e^{i\pi/6}$.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: De Moivre's Theorem for Powers

Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64 ✓

b. -64

c. $64i$

d. $-64i$

In polar form, $1 + i\sqrt{3} = 2e^{i\pi/3}$. By De Moivre's Theorem: $(2e^{i\pi/3})^6 = 2^6 e^{i6(\pi/3)} = 64e^{i2\pi} = 64(\cos 2\pi + i\sin 2\pi) = 64(1 + 0) = 64$.

25 [Foundational] • Target: De Moivre's Theorem for Powers

Evaluate $(\cos(\pi/5) + i\sin(\pi/5))^{10}$.

a. 1 ✓

b. -1

c. i

d. $-i$

By De Moivre's Theorem: $(\text{cis}(\pi/5))^{10} = \text{cis}(10 \times \pi/5) = \text{cis}(2\pi) = \cos(2\pi) + i\sin(2\pi) = 1$.

26 [Foundational] • Target: De Moivre's Theorem for Powers

Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$ ✓

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

Expand $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. Equating real parts: $\cos(3\theta) = \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

27 [Foundational] • Target: Roots of Complex Numbers

Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$ ✓

b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2,\ -1 + i\sqrt{3},\ -1 - i\sqrt{3}$

$8i = 8e^{i\pi/2}$. The cube roots have magnitude $8^{1/3} = 2$ and arguments $\frac{\pi/2 + 2k\pi}{3} = \frac{\pi}{6} + \frac{2k\pi}{3}$ for $k = 0, 1, 2$, giving $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6} \equiv -\frac{\pi}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Roots of Complex Numbers

How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?

a. They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians ✓

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

All nth roots share the identical modulus $r^{1/n}$ and their arguments differ by constant increments of $\frac{2\pi}{n}$, forming a regular n-sided polygon centered at the origin.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Roots of Complex Numbers

What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0 ✓ **b.** 1

c. n **d.** z

By Vieta's formulas applied to $w^n - z = 0$, the coefficient of w^{n-1} is zero, so the sum of all roots is strictly zero.

30 [Intermediate] • Target: Roots of Unity & Factorization

If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0 ✓ **b.** 1

c. -1 **d.** 3

For any nth root of unity $\omega \neq 1$, $\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0$. For $n = 3$, $1 + \omega + \omega^2 = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Roots of Unity & Factorization

Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$ ✓

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

Complete the square: $z^4 + 1 = (z^2 + 1)^2 - 2z^2 = (z^2 + 1)^2 - (\sqrt{2}z)^2 = (z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$.

32 [Intermediate] • Target: Roots of Unity & Factorization

The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. **A square inscribed in the unit circle with vertices at $\pm 1, \pm i$** ✓

b. An equilateral triangle

c. A regular hexagon

d. A rectangle of perimeter 4

The solutions to $z^4 = 1$ are $1, i, -1, -i$. Plotted on the Argand diagram, they form a square of side length $\sqrt{2}$ inscribed in the unit circle.

33 [Intermediate] • Target: Polar and Euler Forms

Find all cube roots of unity.

a. $1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2}$ ✓

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

$z^3 = 1 \Rightarrow z_k = e^{2\pi ik/3}$ for $k = 0, 1, 2$: $1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: De Moivre's Theorem for Powers

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

a. **4 - 6i** ✓

b. 6 + 6i

c. 4 + 6i

d. 6 - 6i

We are asked to find the difference between two complex numbers, z_1 and z_2. Given $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. We need to calculate $z_1 - z_2$. $z_1 - z_2 = (5 - 2i) - (1 + 4i)$ To subtract complex numbers, we subtract the real parts and the imaginary parts separately. Real part: $5 - 1 = 4$ Imaginary part: $(-2i) - (4i) = -2i - 4i = -6i$ Therefore, $z_1 - z_2 = 4 - 6i$. This matches choice A.

35 [Intermediate] • Target: Roots of Complex Numbers

Simplify $\frac{19 - 9i}{-3 + 5i}$, giving your answer in the form $x + yi$.

a. $-3 - 2i$ ✓

b. -4

c. -1

d. -5

Working through this complex-numbers step: $\frac{19 - 9i}{-3 + 5i} \times \frac{-3 - 5i}{-3 - 5i} = -3 - 2i$

36 [Intermediate] • Target: Roots of Unity & Factorization

Simplify $(4 + 2i)(-5 - 5i)$, giving your answer in the form $x + yi$.

a. $-6 - 33i$

b. $-8 - 33i$

c. $-12 - 33i$

d. $-10 - 30i$ ✓

Working through this complex-numbers step: $(4 + 2i)(-5 - 5i) = 4(-5) + 4(-5)i + 2(-5)i + 2(-5)i^2 = -10 - 30i$

37 [Intermediate] • Target: Polar and Euler Forms

Simplify $\frac{12 - 14i}{4 + 2i}$, giving your answer in the form $x + yi$.

a. $-3i$

b. $-6i$

c. $1 - 4i$ ✓

d. $-2i$

Working through this complex-numbers step: $\frac{12 - 14i}{4 + 2i} \times \frac{4 - 2i}{4 - 2i} = 1 - 4i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 1.04 Partial Fractions

Simplify $\frac{18+i}{4+3i}$, giving your answer in the form $x + yi$.

a. 1

b. $4 - 4i$

c. $3 - 2i$ ✓

d. $4 - 3i$

Working through this complex-numbers step: $\frac{18+i}{4+3i} \times \frac{4-3i}{4-3i} = 3 - 2i$

39 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

For $x^2 - x - 56 = 0$, find the product of the roots.

a. -42

b. -47

c. **-56** ✓

d. -51

Working through this complex-numbers step: Product of roots $= \frac{c}{a} = -56 = -56$

40 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$ ✓

c. $-1 + 12i$

d. $-3 + 14i$

Working through this complex-numbers step: $(1 + 2i)(5 + 5i) = (5) + (5)i + 2(5)i + 2(5)i^2 = -5 + 15i$

41 [Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 3.01 3D Vectors & the Cross Product

For $x^2 - 12x + 27 = 0$, find the product of the roots.

a. 35

b. **27** ✓

c. 39

d. 19

Working through this complex-numbers step: Product of roots $= \frac{c}{a} = 27 = 27$

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

For $x^2 - 11x + 18 = 0$, find the product of the roots.

a. 26

b. 19

c. **18** ✓

d. 25

Working through this complex-numbers step: $\text{Product of roots} = \frac{c}{a} = 18 = 18$

43 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

For $x^2 - 6x + 5 = 0$, find the sum of the roots.

a. **6** ✓

b. 11

c. 5

d. 3

Working through this complex-numbers step: $\text{Sum of roots} = -\frac{b}{a} = -(-6) = 6$

44 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

For $x^2 + 3x - 28 = 0$, find the sum of the roots.

a. -1

b. **-3** ✓

c. -6

d. -7

Working through this complex-numbers step: $\text{Sum of roots} = -\frac{b}{a} = -3 = -3$

45 [Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 4.02 The Poisson Distribution

Write $z = 2(\cos 30^\circ + i\sin 30^\circ)$ in the form $x + yi$ (exact values).

a. $\sqrt{3} - i$

b. $\sqrt{3} + i$ ✓

c. $1 + \sqrt{3}i$

d. $-\sqrt{3} + i$

Working through this complex-numbers step: $z = 2\cos 30^\circ + 2\sin 30^\circ\, i = \sqrt{3} + i$

© ClipSAT — clipsat.org | All Rights Reserved